

1. Administrative & Study Identification

Full Study Title: Substudy 06C: A Study of Sacituzumab Tirumotecan (MK-2870) With Pembrolizumab (MK-3475) and Chemotherapy in Participants With First-Line Locally Advanced Unresectable/Metastatic Gastroesophageal Adenocarcinoma (MK-3475-06C/KEYMAKER-U06) **Short Title/Acronym:** KEYMAKER-06C / MK-3475-06C **Protocol Number:** 3475-06C **NCT Number:** NCT06469944 **Posting ID:** 525113344818553890 **Trial Status:** RECRUITING **Sponsor/Company Name:** Merck (Merck Sharp & Dohme LLC) **Investigational Product:** Sacituzumab Tirumotecan (MK-2870, TROP2 ADC) + Pembrolizumab (MK-3475) + Chemotherapy (Capecitabine, Oxaliplatin, 5-FU, Leucovorin/Levoleucovorin) **Phase of Development:** PHASE1, PHASE2 **Medical Monitor:** Medical Director, Merck Sharp & Dohme LLC

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety, tolerability, and establish a Recommended Phase 2 Dose (RP2D) during the safety lead-in phase, measuring Dose-Limiting Toxicities (DLTs), general Adverse Events (AEs), and AE-related discontinuations. In the efficacy phase, to evaluate Objective Response Rate (ORR) per RECIST 1.1 by Blinded Independent Central Review (BICR). **Secondary Objectives:** To evaluate Progression-Free Survival (PFS), Duration of Response (DOR), Overall Survival (OS), and the incidence of Antidrug Antibodies (ADA) to the investigational agents.

2.2 Background & Rationale Treatment options for patients with gastroesophageal adenocarcinoma (GEA) are limited and the overall prognosis is poor. This umbrella platform substudy evaluates the novel combination of a TROP2 antibody-drug conjugate (sacituzumab tirumotecan) with a PD-1 inhibitor (pembrolizumab) and standard chemotherapy as a first-line (1L) therapy, aiming to enhance the immune system's ability to fight the tumor and improve clinical outcomes for patients with advanced or metastatic HER2-negative GEA.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Umbrella Platform Design) **Control Type:** Single-arm (for this specific Substudy 06C) **Blinding:** Open-label **Randomization:** N/A (Dose escalation/expansion)

3.2 Planned Enrollment & Duration Targeted Enrollment: 130 participants worldwide **Number of Sites:** Multicenter, global (including US and Switzerland) **Estimated Study Start:** 19-Sep-2024 **Estimated Primary Completion:** 12-Sep-2029

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Histologically and/or cytologically confirmed diagnosis of previously untreated locally advanced unresectable or metastatic 1L gastroesophageal adenocarcinoma (gastric, gastroesophageal junction, or esophageal). 2. Tumor tissue confirmed as negative for human epidermal growth factor receptor 2 (HER2) expression as classified by ASCO/CAP guidelines. 3. Has measurable disease per Response Evaluation Criteria in Solid Tumors Version 1.1 (RECIST 1.1) as determined by local site investigator/radiology and verified by BICR. 4. Has an Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1 assessed within 3 days before allocation. 5. Has a life expectancy of at least 6 months, and is not expected to require tumor resection during the treatment course. 6. HIV-infected participants must have well-controlled HIV on ART; HBV/HCV infected participants are eligible under strict, controlled viral load/therapy criteria.

4.2 Exclusion Criteria 1. Has squamous cell or undifferentiated gastroesophageal cancer. 2. Has had previous systemic therapy for locally advanced unresectable or metastatic gastric/GEJ/esophageal adenocarcinoma, or has received prior TR0P2-targeted ADC or Topoisomerase I inhibitor-based ADC/chemotherapy. 3. Has experienced weight loss >20% over 3 months before the first dose of study intervention. 4. Has a history of documented severe dry eye syndrome, severe Meibomian gland disease and/or blepharitis, or severe corneal disease. 5. Has Grade >2 peripheral neuropathy. 6. Has active inflammatory bowel disease (IBD) requiring immunosuppressants, or an active autoimmune disease requiring systemic treatment in the past 2 years. 7. Has uncontrolled, significant cardiovascular or cerebrovascular disease within 6 months prior, or accumulation of pleural, ascitic, or pericardial fluid requiring drainage within 2 weeks prior. 8. Known active central nervous system (CNS) metastases and/or carcinomatous meningitis. 9. Has GI obstruction, poor oral intake, or poorly controlled diarrhea.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: - Pembrolizumab administered on Day 1 of each 6-week cycle. - Sacituzumab Tirumotecan (MK-2870) administered on Days 1, 15, and 29 of each 6-week cycle. - Chemotherapy: Capecitabine (Trade Name: Xeloda, ATC Code: L01BC06) twice daily (Days 1-14 & 22-35) OR Oxaliplatin/5-FU/Leucovorin on specific interval days of the 6-

week cycle. [EMA Product Information for Capecitabine available via: https://www.ema.europa.eu/en/documents/product-information/xeloda-epar-product-information_en.pdf] **Route of Administration:** Intravenous (IV) for Pembrolizumab, MK-2870, Oxaliplatin, Leucovorin, and 5-FU; Oral (PO) for Capecitabine.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for chemotherapy-induced side effects (e.g., antiemetics). **Prohibited:** Systemic steroids for autoimmune conditions or immunosuppressive modulating medications that interfere with PD-1 checkpoint inhibition.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Biopsy (HER2 status), Baseline Imaging
Baseline	Day 1 (Cycle 1)	First Dose of Pembrolizumab, MK-2870, and Chemotherapy, Safety Evaluation
Interim	Q6W (Every 6 Weeks)	Days 1, 15, and 29 infusions; tumor imaging for response; AE/DLT assessments
End of Study	Follow-up	Final Vital Signs, Exit Interview, Safety and Overall Survival (OS)
Follow-up		

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints (Safety Lead-in Phase)

- Number of Participants Who Experience One or More Dose-Limiting Toxicities (DLTs):** Defined as any drug-related AE according to NCI CTCAE Version 5.0 observed during the DLT evaluation period.
- Number of Participants Who Experienced an Adverse Event (AE):** Any untoward medical occurrence in a clinical study participant, temporally associated with the use of study intervention.
- Number of Participants Who Discontinued Study Intervention Due to an AE.**

7.2 Secondary & Exploratory Endpoints

- Progression-Free Survival (PFS) per RECIST 1.1 as Assessed by BICR.
- Duration of Response (DOR) Per RECIST 1.1 as Assessed by BICR.
- Overall Survival (OS).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of treatment-emergent AEs; specific focus on establishing the RP2D during the initial safety lead-in phase using a Bayesian optimal interval design. **Serious Adverse Events (SAEs):** Reported promptly according to standard oncologic protocol guidelines, including monitoring for severe hypersensitivity and immune-mediated adverse reactions. **Data Monitoring Committee (DMC):** Internal

Safety Review Committee evaluating DLTs during the safety lead-in cohort before expansion.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: - *Safety Set:* All participants who receive at least one dose of the study intervention. - *Efficacy Set:* All enrolled patients with measurable disease evaluated for ORR.

Sample Size Justification: A total of 130 patients will be enrolled to evaluate the efficacy profile adequately, with early evaluation to define the RP2D. **Handling of Missing Data:** Standard oncological trial imputation for missing tumor assessments and survival follow-ups.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: High-frequency onsite monitoring during the safety lead-in phase for DLTs, followed by regular data verification visits. **Audit Trail:** Secure eCRF systems, imaging repositories for BICR, and comprehensive clinical data clarification forms.

11. Study Identifiers & Governance

Primary ID: 3475-06C / **MK-3475-06C Registry Number:** NCT06469944

Posting ID: 525113344818553890 **Sponsor/Lead Organization:** Merck (Merck Sharp & Dohme LLC)

Study Title: KEYMAKER-U06 Substudy 06C
Official Regulatory Title: A Phase 1/2 Open-Label, Umbrella Platform Design Study of MK-2870 With Pembrolizumab (MK-3475) and Chemotherapy in Participants With 1L Locally Advanced Unresectable/Metastatic Gastroesophageal Adenocarcinoma (Gastric Adenocarcinoma, Gastroesophageal Junction Adenocarcinoma, and Esophageal Adenocarcinoma): Substudy 06C

12. Clinical Framework (Gastroesophageal Adenocarcinoma Specific)

Disease Areas: Gastroesophageal Junction, Gastroesophageal Adenocarcinoma, Esophageal Neoplasms, Esophageal Cancer **Primary Condition / Preferred Name:** Adenocarcinoma (UMLS Preferred Name: Gastroesophageal Adenocarcinoma) **Diagnosis Threshold:**

Histologically/cytologically confirmed HER2-negative disease

Medical Methodology: TROP2-targeted Antibody-Drug Conjugate + PD-1 Checkpoint Inhibitor + standard Chemotherapy **Optimization**

Target: Maximizing tumor response (ORR/PFS) safely without compounding overlapping toxicities

13. Study Procedures & Methodology

Management Pathway: Master Umbrella Protocol assignment (KEYMAKER-U06) branching into targeted substudies based on biomarker (HER2) status. **Education Protocol (HCP):** Site initiation and continuous training regarding infusion protocols, DLT criteria, and identification/management of immune-mediated AEs (e.g., pneumonitis, severe dry eye). **Education Protocol (Patient):** Strict guidelines and diaries for proper at-home administration of oral chemotherapy (capecitabine/Xeloda) and reporting of treatment-emergent symptoms. **Quality Audit Mechanism:** Continuous monitoring of Blinded Independent Central Review (BICR) concordance with investigator assessments for RECIST 1.1.

14. Observation & Follow-Up Schedule

Total Duration: ~5 years total duration (Primary completion estimated Sep 2029) **Onsite Visit Cadence:** Frequent cycle visits (Days 1, 15, 29 of 6-week cycles) **Remote Monitoring Frequency:** Intermittent telehealth and symptom checking between infusion dates **Checklist Review Interval:** End of every 6-week cycle aligned with scheduled restaging imaging

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	03-Apr-2026				
Clinical Operations Lead	[Name]	_____	03-Apr-2026		Lead		
Statistician	[Name]	_____	03-Apr-2026				